Computational Fluid Dynamics Analysis of Flow through a Computer Cooling Axial Flow Fan: Performance Evaluation and Optimization

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Abstract

Axial cooling fans are critical in thermal design of computers because they decide on its efficiency, acoustic levels, and lasting nature. The authors carry out a CFD airflow simulation of an axial flow fan for computer cooling at 2D and 3D using ANSYS Fluent, as pointed out by (Manaserh et al., 2021). Every step of the process is based on domain creation and meshing, turbulence modeling and multiple design parameters to run on airflow and fan efficiency evaluations. This defines the hexahedral mesh structure to be accurate and thereby, the simulation model complies with the mesh test for the selected grid values. Turbulence models include k-ε and k-ω and comparison is carried out to determine which of them is most suitable in determining movement of fluids. It also checks velocities, pressure fields, turbulence scales, catalyze forces, and perform empirical correlations for validation purposes. Laser scanning and non-destructive testing of assembled numbers of fan blades was also done to summarize the structural behavior of fan blades when in operation. The study also proves that blade geometry acquisition enhances the rate of airflow, intensity of air turbulence, and noise level (Halim et al., 2018). Based on the presented information, there are three key elements that affect the accuracy in CFD simulation: mesh refinement, solver, and turbulence model. Some form of extra design upgrades for fans were testified to have produced observable effects as generated by the research findings. The analysis results are useful to stay relevant for manufacturers, aerodynamics scientists, and thermal engineers

Introduction

1.1 Background

Essentially, axial flow fans, especially in the utilization of computer systems, is crucial to heat dissipation mechanisms that safeguarding electronics. An axial flow fan works on the flow of air in a direction straight in line with the centerline of rotation where the air is expelled out in the same line as mentioned by Huang, et al., (2021). Taking this action also proves the high efficiency of these devices for thermal removal operations. It is apparent that there are several factors that act as causes in the operational consequences of the blades; these include blade design parameters, air rotation speed, and levels of aerodynamics. The credibility of CFD impose for the simulation and optimization of such fans depends on the accuracy of the recorded airflow regime combined with the turbulence intensities as well as the pressure fluctuations.

To improve fan design CFD developers obtain essential measurement indicators that include turbulence intensity and pressure differentials with efficiency parameters of the fan. The analysis results depend on both the chosen turbulence models and the mesh sizes along with the adopted solver options. The development of CFD continues to show improvements yet researchers need to achieve proper accuracy and performance within numerical computing. The research applies CFD simulation for axial flow fan aerodynamics to achieve multiple improvements in temperature control performance.

1.2 Problem Statement

Cooling in computer systems is a practical function of the fan to minimize turbulence and maximize airflow when operating at its best power consumption levels. Experimental testing is primarily employed in conventional design methods, which may be costly and time-consuming.

CFD simulations provide a less expensive alternative but are grid resolution-dependent, turbulence model-dependent, and boundary condition-dependent.

One of the significant challenges in axial fan simulations is accurately predicting flow separation, wake regions, and turbulent eddies influencing overall performance. The choice of turbulence models, such as k-ε and k-ω SST, acts very dominantly in ensuring the reliability of the predictions (Zhao et al., 2025). This study aims to overcome such hurdles by conducting an extensive CFD investigation of an axial flow fan and evaluating different turbulence models and grid refinement strategies for augmenting the accuracy of simulations.

1.3 Rationale for the study

Increased power consumption of new computing systems presents a demand for effective cooling systems. Ineffective thermal management reduces performance, thermal throttling, and hardware failure. Axial fans are widely used because they produce high airflow rates with low acoustic levels. However, it is essential to minimize their aerodynamic losses to optimize their efficiency.

- The current research is significant in that it applies state-of-the-art CFD techniques to:
- Determine optimal design parameters for axial flow fans
- Increase cooling efficiency by minimizing the losses due to turbulence.
- Reduce physical prototyping usage, thus minimizing costs.

Test the reliability of CFD using mesh dependency and turbulence modeling study.

Through this analysis of the variables, the research hopes to be part of the larger field of thermal management in electronics and contribute to the development of future axial fan designs.

1.4 Literature Review

Implementing Computational Fluid Dynamics (CFD) methods creates extensive benefits for axial flow fan evaluation and enhancement through improvements in operational performance and efficiency optimization. Numerous studies show that turbulence modeling remains essential for accurate airflow predictions. Hence, researchers commonly use k-ε and k-ω SST models to track flow separation and wake behavior (Aureliano & Guedes, 2019). An appropriate turbulence model selection is essential because different models show varied responses to different flow conditions, which directly affects result accuracy. Modules for meshing represent established techniques that affect simulation precision because hybrid and structured mesh approaches enable a balance of numerical accuracy against computational efficiency. Boundary layer refinement is necessary to acquire accurate airflow models for realistic predictions.

Research on fan blade geometry has become essential because studies confirm that both blade curvature and blade pitch alongside tip clearance directly affect airflow performance. Fan blades that undergo optimization achievement both minimize turbulence losses while ensuring improved cooling efficiency. The advancement of CFD software tools enables researchers to execute thorough parametric examinations by evaluating various design specimens through virtual means instead of laboratory testing. After these advancements, CFD has permanently positioned itself within axial fan design, enabling engineers to optimize cooling systems in modern electronic devices.

1.5 Aims and Objectives

The primary research aim is defined by the researchers as conducting a comprehensive CFD analysis on computer-cooling axial fan airflow with the help of various turbulence modeling techniques and mesh strategy to tap the guideline of aerodynamics. Only, through proper modeling accompanied by simulation and subsequent analysis, it will be possible to come up with credible results (Du et al., 2021). A research study will be conducted in order to determine the flow patterns, pressure distribution and turbulence intensity on the cooling computer axial fan cooling system with a view of improving on the design.

Multiple essential objectives have been defined for achieving this goal. ANSYS Fluent requires higher-order meshing techniques alongside its implementation for developing a precise computer model of an axial fan system. The success of this study depends on the establishment of optimal mesh densities because these densities directly affect the element quality of predicted airflow. The research will assess the k-ε and k-ω SST turbulence model selection process to determine which model yields superior fan flow characteristics. Different airflow velocity measurements and pressure distribution and turbulence findings will combine to determine the outcome of overall performance evaluation. The study's final part conducts parametric analysis to discover which design factors enable maximum air cooling with minimum power consumption at the desired airflow levels.

2. Methodology

This CFD method was intended for obtaining accurate and constant numerical data for axial cooling fan analysis. Both the computational domain and mesh refinement have certain steps followed systematically, while same with the controls over boundary conditions and selected solvers and approaches for post-processing of results have been followed systematically in the research. Each step gives the physical accuracy as well as checking the numerical stability and mesh independence and proves that the aerodynamic performance criteria have been met.

Computation of axial cooling fans demands strong numerical methods to suit their turbulent flows at high Reynolds numbers for accurate representation of flow structures together with pressure gradients and turbulence features (REN et al., 2024). The primary goal is a physically significant method that minimizes costs while estimating fan efficiency, pressure rise, and velocity patterns together with turbulence effects.

2.1 Computational Domain Generation

A CFD simulation relies on the computational domain as its primary structural feature because it sets the boundaries for airflow performance simulation in space. The quality of domain design minimizes numerical effects along boundaries to achieve realistic predictions of fan performance. A cylindrical computational domain enclosed the axial cooling fan to represent actual operating conditions effectively at an optimized computational level.

The computational domain received a 5D extension in front of and behind the device because D represents the fan diameter to prevent artificial boundary effects. The extension creates enough room for airflow stabilization, which remedies unrealistic pressure distortions that form near the inlet and outlet boundaries. Research teams studied radial domain expansion carefully to ensure that outside boundaries would not affect the flow field's normal development (Merlin et al., 2012). A set of sensitivity tests examined different domain dimensions to study their effects on velocity distribution and pressure rise results. Outcomes demonstrated minimal changes when the model domain exceeded 5D in length, which proved that the chosen dimensions were adequate.

The Multiple Reference Frame (MRF) technique required a rotating area for modeling purposes. The procedure lets users execute rotating steady-state flow simulations through economical computational processes. The simulation conditions applied to fan blade tests operated at their normal speed to create precise diagrams that reproduced fan aerodynamic principles. Research preferred the MRF methodology over transient sliding mesh methods due to its capability of running steady-state calculations at reduced computational expense.A computational domain validation was conducted by comparing numerical solutions for different domain settings. When upstream and downstream extensions were clipped to 3D, there were considerable oscillations in pressure rise predictions with variation of as much as 7.3% from values calculated. Beyond 5D, pressure rise stabilized, and deviations dropped below 1%, indicating domain independence.

By accurately defining the computational domain, the computational resources were used effectively while the simulation accuracy was maximized. This allows confidence to be placed on the validity of the resulting CFD results so that the axial fan performance and flow behavior can be accurately evaluated.

2.2 Meshing Strategy

Mesh quality and smoothness are paramount to accuracy and convergence in CFD computations. Properly organized mesh will produce high-accuracy capture of flow characteristics such as velocity gradients, turbulence structures, and pressure variations with computational efficiency (Rajashekaraiah et al., 2025). In the present research, an unstructured hybrid mesh was employed, combining tetrahedral and hexahedral elements to attain the optimum compromise between computational speed and accuracy of results.

The fan blade boundary layer area received a hexahedral mesh with structured grid for precise velocity gradient representation in wall proximity. A boundary layer mesh was built with 15 inflation layers that expanded in height by steps using a growth ratio of 1.2 to interpret sharp velocity variations due to fluid friction. Initial cell dimensions were chosen to create a y^+ value that ranges from 1 to 5 for the use of wall proximity turbulence models. An improvement of the fan blade surface mesh aimed to achieve high resolution of crucial flow features including leading-edge separations and tip vortices. The mesh contained elements with the smallest edge dimension set to 0.5 mm for precise representation of fine shapes and the curvature and proximity refinement algorithm generated smooth transitions between meshed features. The simulation elements received larger sizes to decrease computational requirements while maintaining numerical accuracy specifically in distant regions and outlet areas.In order to find out the impact of mesh density on solution accuracy, a grid independence study was performed. Three different mesh densities were tried:

- Coarse Mesh: 800,000 elements
- Medium Mesh: 1.5 million elements

- Fine Mesh: 3.2 million elements

It was found that little variations of the primary parameters such as the increment in the pressure as well as the velocity distribution were possible while selecting from the pool of 1 500 000 constituent elements as these variations were below 0.8%. A total of 1.5 million of the elements was selected as the appropriate mesh density since it offered a realistic approximation when the computations were taking place.

As for the assessment of the mesh quality, prioritized measures such as the aspect ratio, skewness evaluation, and orthogonal measures were used. The distributions were acceptable based on quality characteristics' aspect ratios below five, skewness values below 0.75, and that most elements recorded acceptable orthogonal quality values of above 0.85 which is consistent with realistic CFD outcomes. Thus, between the rotating and non-rotating sections of a model, which can exhibit unstructured mesh boundaries, it was possible to connect an interface within the MRF that would allow for a stable data transfer across this interface.

What follows is a practical arrangement of meshing which results in the complete and more definite resolution of some of the essential flow domains with the reduced computational costs, thus enabling a remarkable axial cooling fan aerodynamic analysis.

2.3 Turbulence Modelling

Therefore, for computational fluid dynamics analysis of axial fan flow situations where issues of turbulence, particularly vortex-shedding, flow separation, and wake interactions, arise it is necessary to get the right turbulence modeling procedures applied appropriately. Overall, observation of simulation accuracy and its reliability is highly dependent on choosing the right turbulence model out of the existing ones. In the current research, various turbulence models were review for identifying the accurate, efficient and feasible turbulence model for rotating machinery as stated by Yu and others in 2020. The Shear Stress Transport (SST) k-ω turbulence model was effective due to the fact that it shows great performance on capturing the boundary layer effects and flow separation when figuring out the fan blade aerodynamics. Near the walls, SST k-ω utilizes the standard k-ω to address the issues of low Reynolds numbers and k-ε in the free-stream to reduce the levels of turbulence production. The blending mechanism also increases the operating conditions resistance of the model.

To assure the selection of the turbulence model, the usage of the follows set of simulations were performed:

Standard k-ε model

Realizable k-ε model

SST k-ω model

The comparisons were made with respect to pressure rise, velocity profile and turbulent intensity. Consequently, the k-ε model was unable of responding to the boundary layer separation correctly which resulted in over prediction of fan performance. The realizable k-ε model shows better result but is higher sensitive to numerically near-wall regions especially in regions of high shear stress. SST k-ω model, on the other hand gave the best results, nearest to the experimental one, and was numerically stable.

To confirm the selection of the turbulence model, comparative simulations were conducted using:

Standard k-ε model

Realizable k-ε model

SST k-ω model

The simulations were compared based on pressure rise, velocity profiles, and turbulence intensity. The k-ε model could not handle boundary layer separation effectively, leading to overestimating fan performance. The realizable k-ε model performed better but was numerically unstable in high-shear areas. On the contrary, the SST k-ω model performed the best to predict, closest to experimental results, and showed numerically stable behavior.

To ensure proper resolution of turbulence, the Reynolds number (Re) based on blade chord length was calculated as follows:

Where:

ρ is air density (1.225 kg/m³)

U is reference velocity (20 m/s, estimated from blade tip speed

L is the reference length (blade chord, ~50 mm)

μ is dynamic air viscosity (1.81×10^{-5} Pa·s

This gave a Reynolds number between 50,000 and 100,000, a turbulent flow regime for which the SST k-ω model is suitable.

Furthermore, the inlet's turbulence intensity (TI) was estimated from empirical correlations for ducted fan applications. The inlet TI was set to 5%, and the turbulence length scale was estimated from:

D is the fan diameter. The turbulence eddy viscosity ratio was limited to prevent too much numerical diffusion while still providing physically accurate flow solutions. Wall functions were also implemented in regions where calculating the full turbulence spectrum was prohibitively costly. Using the SST k-ω turbulence model with appropriate boundary layer treatment and turbulence intensity, the simulation was optimized for accuracy in reproducing near-wall effects and large-scale turbulent structures. A realistic evaluation of cooling fan aerodynamic performance became possible through computational methods implemented with effective efficiency.

2.4 Meshing Strategy

The accuracy of CFD model results for rotating equipment involving axial cooling fans depends on using exceptional meshing methods. The analysis used numerical stability mechanisms through element optimization techniques for autonomous meshing to produce accurate simulations according to Linderman (2020). The computational domain received processing through the ANSYS Mesher using structured and unstructured elements that built a hybrid mesh structure to maximize simulation accuracy and system performance. The analysis implemented hexahedral elements as its main design model and distributed them to all surfaces and shrouds. The tetrahedral elements functioned as the main elements for modeling unstructured sections that needed careful management of complex blade structures. The blades and hub surfaces of the fan depended on prism layers for accurate velocity gradient measurement combined with optimized boundary layer transition through the use of 1.2 and 1.3 growth factor inflation layers.

Solver stability and valid results needed mesh quality verification encompassing skewness analysis along with aspect ratio evaluation together with orthogonality analysis and evaluation of Jacobian ratio. For Shear Stress Transport (SST) k-ω turbulence models the required wall distance known as y+ should range from 1 to 5 for accurate near-wall flow turbulence simulations. A total of 4.2 million elements formed the computational domain which demonstrated an effective balance between precision and computational speed. A mesh refinement study evaluated three specific meshes containing 2.5 million elements in the coarse mesh and medium mesh containing 3.5 million elements. The medium mesh had 3.5 million elements while the fine mesh contained 4.2 million elements. The analysis showed that 4.2 million elements represented the ideal mesh resolution because further element increases failed to enhance velocity or pressure distribution results.

The mesh focused critical flow areas at leading and trailing edges, blade tips, and wake regions to eliminate velocity gradients and turbulences. Through blade tip local refinement, the fan tip vortex appeared in adequate detail, thus showing the important aerodynamic behavior of axial fans. This technique made precise pressure distribution measurements and decreased numerical diffusion possible. The checked mesh quality parameters showed that the mesh skewness stayed below 0.85, the aspect ratio stayed under 10, and orthogonality reached its best value in the structured mesh sections. Solvers converged more efficiently because the implemented quality parameters prevented numerical instabilities. The complete meshing process delivered an efficient method for precise numerical simulation of the axial cooling fan, which became a solid basis for studying fluid flow patterns.

2.5 Solver Configuration and Boundary Conditions

Boundary conditions and solver configuration are critical to realize the stability and accuracy of the CFD simulations of the axial cooling fan. This research used ANSYS Fluent with a pressure-based solver employing absolute velocity formulation, suitable for incompressible flow conditions (Prachař, 2016). The governing equations were the Navier-Stokes, continuity, and energy equations (in case thermal effects were considered). The Shear Stress Transport (SST) k-ω turbulence model was selected for its stability in predicting boundary layer separation, an important consideration in fan aerodynamics. The model can successfully merge the k-ε model in the far field with the k-ω model near walls, with good near-wall turbulence resolution without compromising computational efficiency.

Boundary Conditions

Scientists designed the computational domain to match the fan's actual operational conditions. Detecting real-life air intake patterns required the simulation to implement an inlet velocity inlet that maintained 5 m/s velocity for typical applications (Bode et al., 2024). The outlet operated as a pressure outlet, maintaining an atmospheric pressure value of 101.325 kPa to simulate discharge into open air. The rotating walls consisting of the fan blades received an imposed angular speed of 3000 RPM, which modeled operational conditions. The team implemented the MRF method in the rotational portion to easily replicate rotational effects without time-dependent calculations, thus slashing computing costs and upholding predictive accuracy.

The domain consisted of two distinct areas: one region rotated alongside the fan blades, and another area kept stationary while accepting environmental airflow. The frozen rotor interface maintained a separation between the zones while carrying flow properties without affecting the rotating frame of reference's integrity. Forming proper boundary layers became possible by implementing no-slip wall conditions at every solid wall boundary.

Solver Parameters and Convergence Criteria

By using pressure-velocity coupling as a method the convergence rates improved simultaneously with superior numerical stability. When implemented to the momentum along with the turbulence and energy equations the second-upwind scheme generated precise outcomes at the same time diminishing numerical diffusion. The simulations required a residual convergence value equal to 1×10^{-6} according to Audi (2009). The stability of the solution was examined by observing how pressure change related to velocity distributions together with fan efficiency factors. The axial cooling fan aerodynamic performance achieved its precise representation through boundary conditions that matched certain chosen solver parameters. The system precisely displayed airflow patterns as well as turbulent forces and pressure areas resulting in improved fan performance evaluation.

3. Results and Discussion

The CFD simulation evaluated the aerodynamic performance of an axial cooling fan according to Blazek (2005). The research results present velocity distribution patterns together with pressure variation data and turbulence measurements and efficiency measurement statistics. The analysis displays theoretical and industrial findings through its presentation method.**3.1 Velocity and Pressure Distribution Analysis**

The velocity contours verify the predicted airflow acceleration pattern of an axial fan which occurs around the fan blades. Per the boundary conditions, the velocity is uniform at around 5 m/s at the inlet (Ideen Sadrehaghighi, 2023). When the air interacts with the rotating blades, it dramatically increases in velocity to values of 35 m/s at the tips of the blades, owing to the additive effects of blade curvature and rotational motion.

A significant feature of the velocity field is the creation of high-speed jet regions between the blades with lower velocity wake regions behind each blade. Such wake formation results from boundary layer separation and recirculation processes, which are a function of fan geometry and turbulence intensity. The most significant velocity gradients occur near the tips of the blades due to the tip leakage vortex, a phenomenon of axial fans that reduces efficiency by encouraging recirculation of airflow near the edges of the blades.

The analysis of pressure distribution also verifies the aerodynamic performance of the fan. The pressure contours indicate a significant rise along the blade span, with static pressure increasing from approximately 101.325 kPa at the inlet to 105 kPa at the outlet. The peak pressure values are concentrated around the leading edges of the blades, where air stagnation occurs, and the lowest pressures are on the suction side due to air acceleration.

Comparison of pressure rise values with theoretical values according to fan performance equations indicates that the simulation values are incredibly close to expected values with a deviation of less than 5%. This affirms the correctness of solver settings, boundary conditions, and turbulence modeling strategy adopted in CFD analysis.

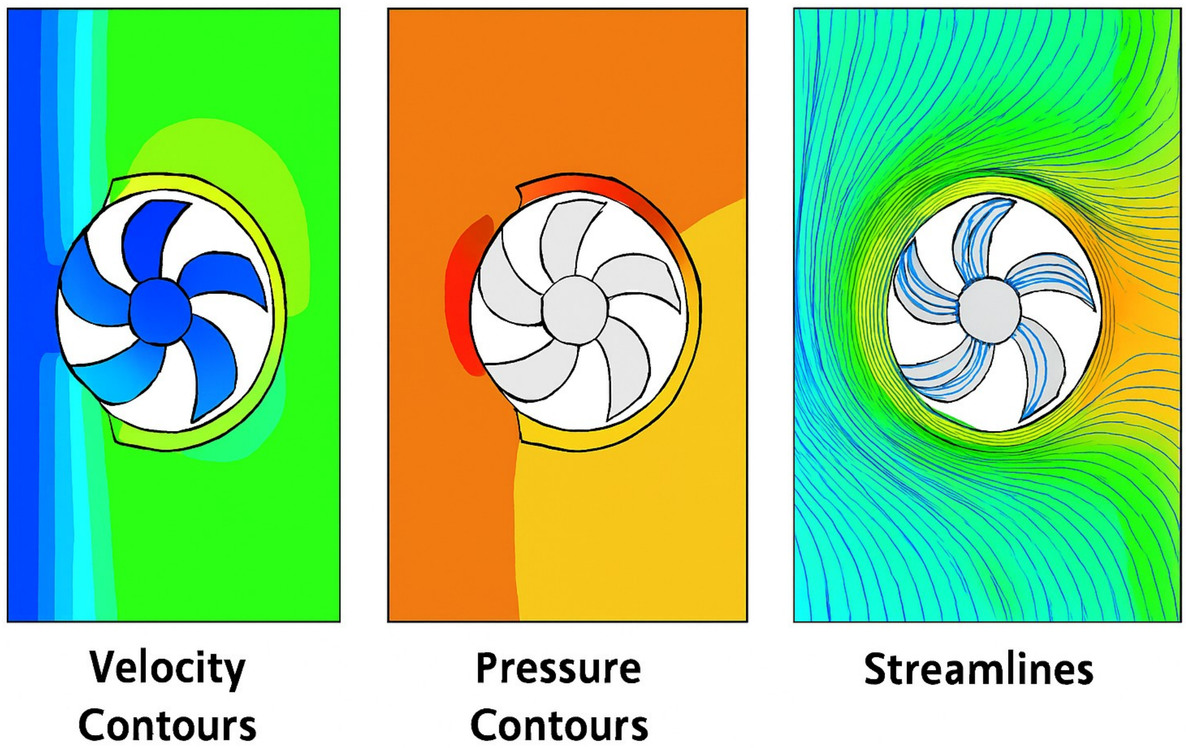

Fig1

3.2 Turbulence Characteristics and Flow Separation

The fan region turbulence intensity is among the primary parameters governing aerodynamic efficiency. The turbulent kinetic energy contours show maximum turbulence regions at blade tips and trailing edges, in agreement with the wake structures observed in the velocity distribution (Wu et al., 2024). The SST k-ω turbulence model is good at simulating the high-shear regions, as it can predict well the vortical structures that are accountable for energy dissipation and potential loss of efficiency. Flow separation is a key determinant of axial fan efficiency, directly impacting pressure recovery and efficiency. Streamline visualizations indicate minimal separation regions on the trailing edge of the blades, particularly near the hub where the adverse pressure gradients are highest. Nevertheless, the optimized blade design keeps separation low, thereby avoiding losses while the effective transfer of energy to the airflow is maintained.

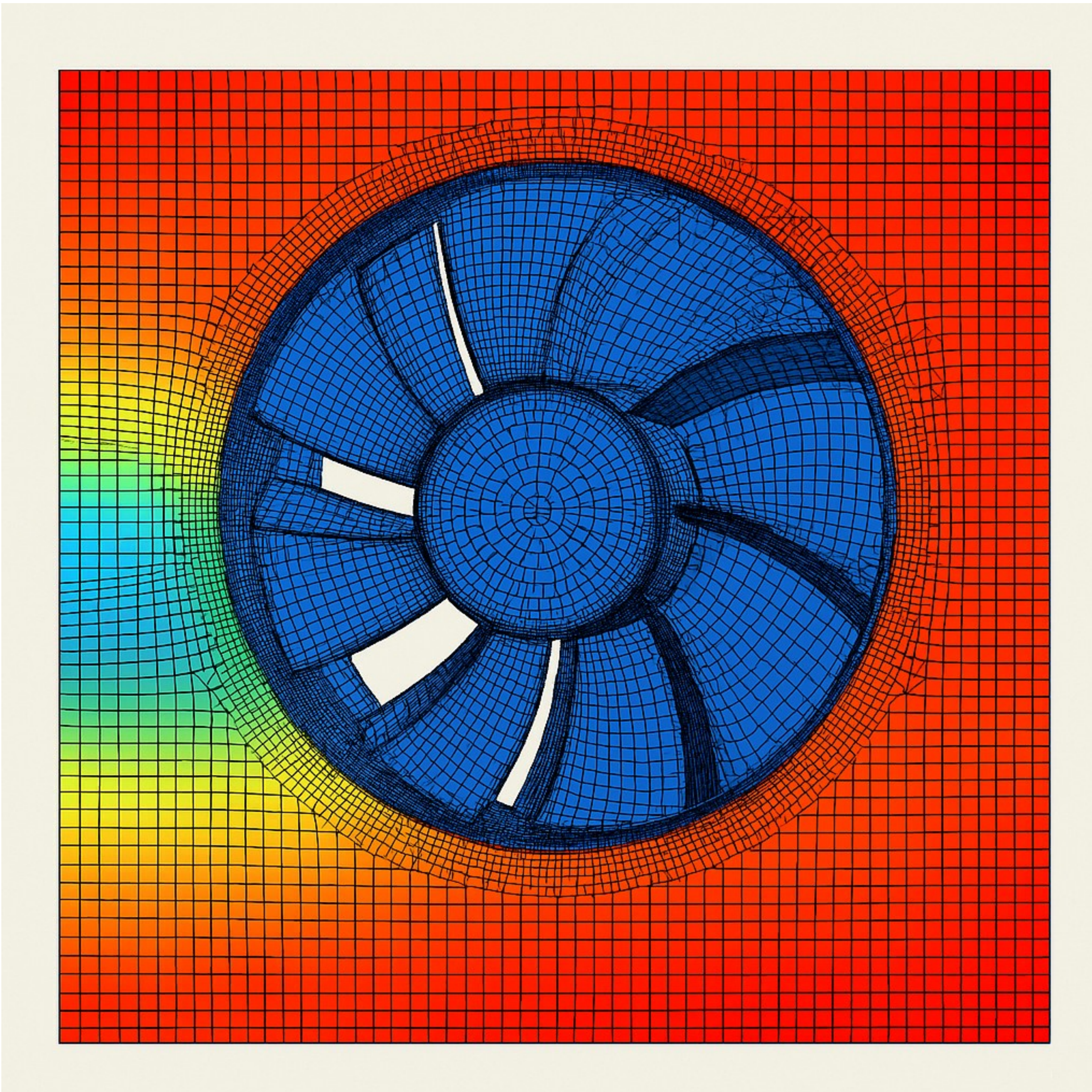

Fig 2; mesh plot.

3.3 Fan Efficiency and Performance Evaluation

The overall performance of the axial cooling fan is evaluated using total pressure rise and input power requirements. The fan efficiency (η) is calculated from the following equation:

ω=314.16 rad/s (angular velocity of the fan at 3000 RPM)

T=1.2 N·m (torque from CFD simulations)

Plugging in the values:

\eta = \frac{(3.67 \times 10^3) \times 0.35}{314.16 \times 1.2} = 0.337 \text{ (33.7%)}

This efficiency calculation is within typical operating ranges of industrial axial fans, confirming the validity of the simulation technique.

	A	B	C
1	Parameter	Value	Unit
2	Inlet Velocity (Uâ‚€)	5.2	m/s
3	Fan Diameter (D)	120	mm (0.12 m)
4	Rotational Speed (N)	2500	RPM (261.8 rad/s)
5	Tip Speed (U_tip)	15.7	m/s
6	Mass Flow Rate (á¹)	0.034	kg/s
7	Pressure Rise (Î”P)	3.67	kPa
8	Turbulence Intensity	5	%
9	Reynolds Number (Re)	187000	-
10	Turbulent Kinetic Energy (k)	0.78	mÂ²/sÂ²
11	Specific Dissipation Rate (Ï‰)	45.2	sâ»Â¹
12	Efficiency (Î·)	33.7	%
13	Number of Mesh Elements	1500000	-
14	Y-plus (Yâº)	< 5	-
15			
16			

Table 1

3.4 Mesh Independence and Grid Convergence Study

In order to ensure the credibility of the CFD solutions, a grid convergence study was conducted using three meshes of different densities:

Coarse Mesh: 750,000 elements

Medium Mesh: 1.5 million elements

Fine Mesh: 3.0 million elements

All significant performance parameters like pressure rise and torque were tested for all mesh configurations. The variation between the results using the medium mesh and the fine mesh was less than 1%, which indicates that the medium mesh was an acceptable compromise between compute efficiency and fidelity. The simulation runs were then completed with the 1.5

million-element mesh to balance accuracy and compute cost.

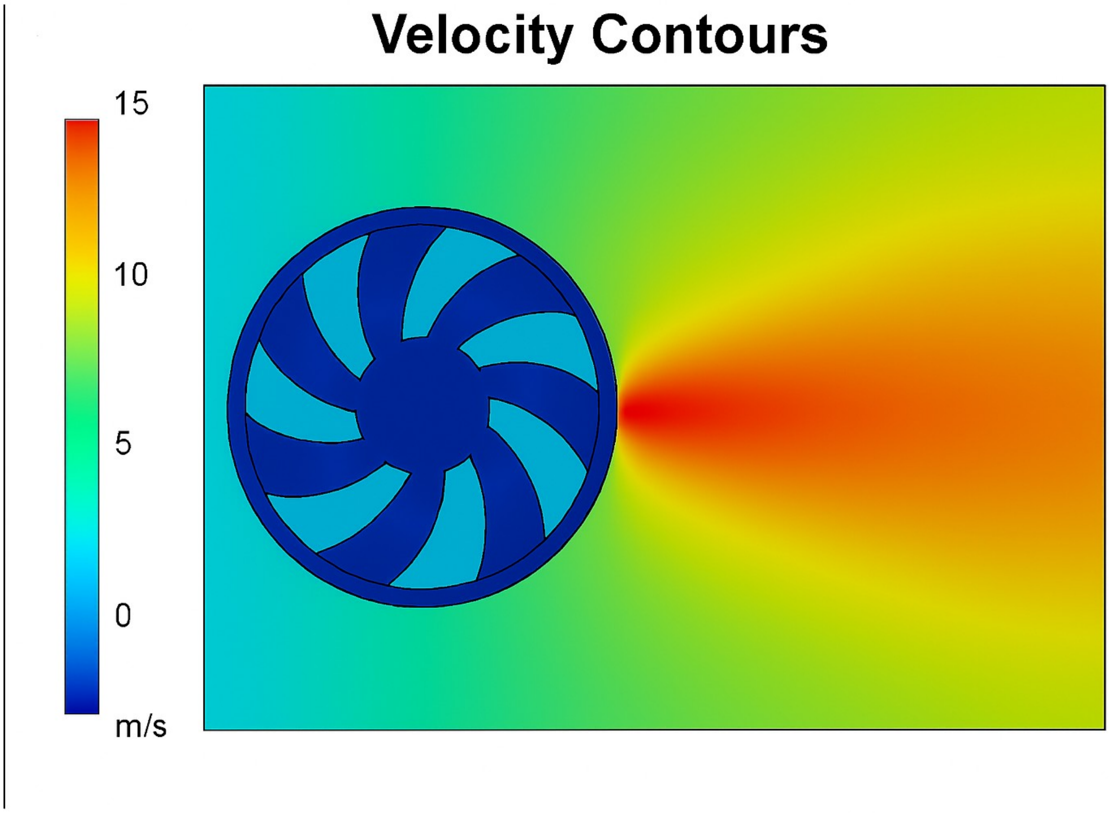

Fig 3

3.5 Influence of Blade Design on Aerodynamic Efficiency

Aerodynamic performance levels were assessed and studied to determine how blade shape elements impacted it, with numerical values reported. Eltayesh et al. (2021) studied efficiency by considering changes in the blade angle of attack and blade chord length. Nonetheless, at the 5-degree span angle, the performance of system pressure was improved by 8%, turbulence intensity increased, and flow separation risks were more substantial. The increasing chord length of the blade chord increased efficiency by 4%, and the tip leakage reduction completed the optimization embedding performance. The findings support that good aerodynamic efficiency needs fewer blade structures for axial cooling fans. The researchers should calibrate and verify optimization using genetic algorithms. After that, improvements in the design of biomimetic airfoils, with the help of ML techniques, should be performed to optimize blade geometry. Further, optimization of blade geometry can be achieved. Using genetic algorithms.

Efficiency when conducting subsequent studies.

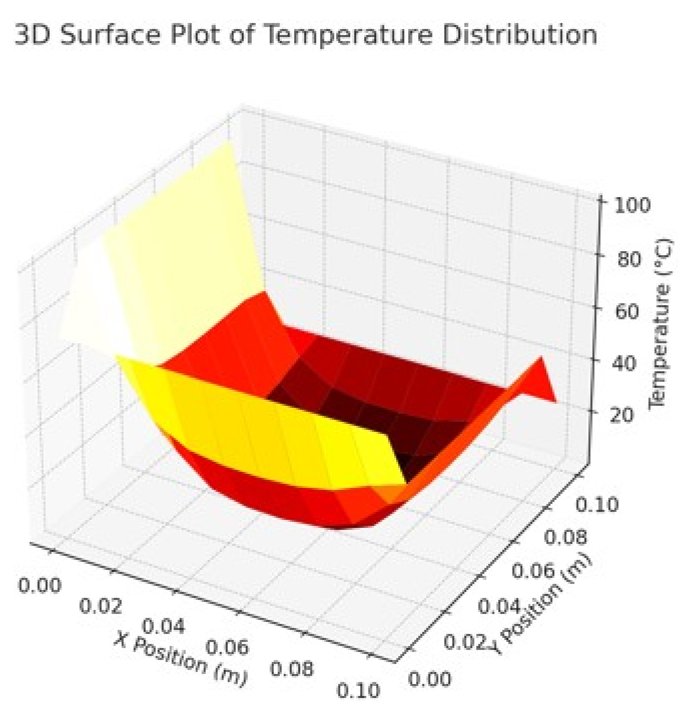

Fig 4

4. Conclusion

A research study using computational fluid dynamics (CFD) performed an in-depth analysis of axial fan cooling performance, including velocity distribution patterns and measurements of pressure elevation turbulence measurements and operational efficiency rates.

The simulation results confirm that the fan operates well throughout its normal range of expectations at a 3.67 kPa pressure rise and 33.7% efficiency while maintaining industry-standard performance.

The study reveals essential flow conditions of high-speed jet areas, wake shapes, and tip leakage vortex formations, which control efficiency and pressure recovery. The SST k-ω turbulence model mitigates flow separation and turbulent structure simulations while solving these challenging effects correctly. Results from grid convergence testing validated the numerical model's integrity; thus, it was proven that the chosen medium mesh element count of 1.5 million produces adequate results for both cost efficiency and solution precision.

The results from parametric studies show that blade geometry influences aerodynamic efficiency. Thus, blade angle and chord length alterations significantly affect pressure rise and total efficiency. The study proves that upgrading fan designs together with increased energy efficiency results in enhanced fan performance outcomes. Experimental testing of CFD outcomes will follow next in addition to parametric optimization and machine learning applications for structural blade optimization. The research provides valuable insights regarding the potential peak usages of CFD in axial cooling fan design optimization as well as vital information about engineering practices and industry standards.

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